# The RedShift

Starting from the onduscular equation particularized for Muon Neutrino which is a Dirac neutrino with spin quantum number s = - ½ thus we have:

$$\frac{R_s}{r}\Delta f(r,t) = \frac{1}{c^2}\frac{\partial^2 f(r,t)}{\partial t^2} \quad (1)$$

The condition that μ travels with the speed of light can be expressed by employing the divergence operator in general coordinates r and c·t thus obtaining the simple equation div[f(r,c·t)]=1, expanded to

$$\frac{df(r,t)}{dt} = \frac{\partial[u(t)g(r)]}{\partial t} + \frac{\partial[u(t)g(r)]}{\partial r}\cdot\frac{dr}{dt} = c. \quad (2)$$

Since the μ neutrino speed will be "c" regardless of the length of the radius r, which means that $\frac{dr}{dt} = c$ for any r, we obtain that for any r and c·t: $g(r)\frac{\partial u(t)}{\partial t} + u(t)\frac{\partial g(r)}{\partial r}\cdot c = c$ (3)

The above condition shows that the space growth rate is equal to "c" all the time.

For the muon neutrino, the flux equation for μ neutrino considering and spherical cords symmetry & the speed of graviton equal to light speed is for (electronic neutrino) Flux second-order equation is:

$$\frac{\partial^2 P(r,t)}{c^2\partial t^2} - \frac{\partial^2 P(r,t)}{\partial r^2} = 0 \quad (4)$$

With initial condition $P(2R_s,t)=0$ si $diff(P(0,0),r)=-R_s$ =-7.061266475 $\cdot 10^{25}$ m (7.463764553billion lightyers);

Thus the solution of Λ Space flux/ surface unit Λflux = P(r,t) /4πr$^2$ equation (5):

$$\frac{P(r,t)}{4\cdot\pi\cdot r^2} = \frac{(_c_2\cdot t^2 + 4\cdot R_s^2 - 2\cdot R_s\cdot r + 2_C3\cdot t)\cdot c^2 + (-4\cdot R_s^2 + r^2)_c_2}{8\cdot\pi\cdot r^2\cdot c^2}$$

And with initial condition t=0 (6):

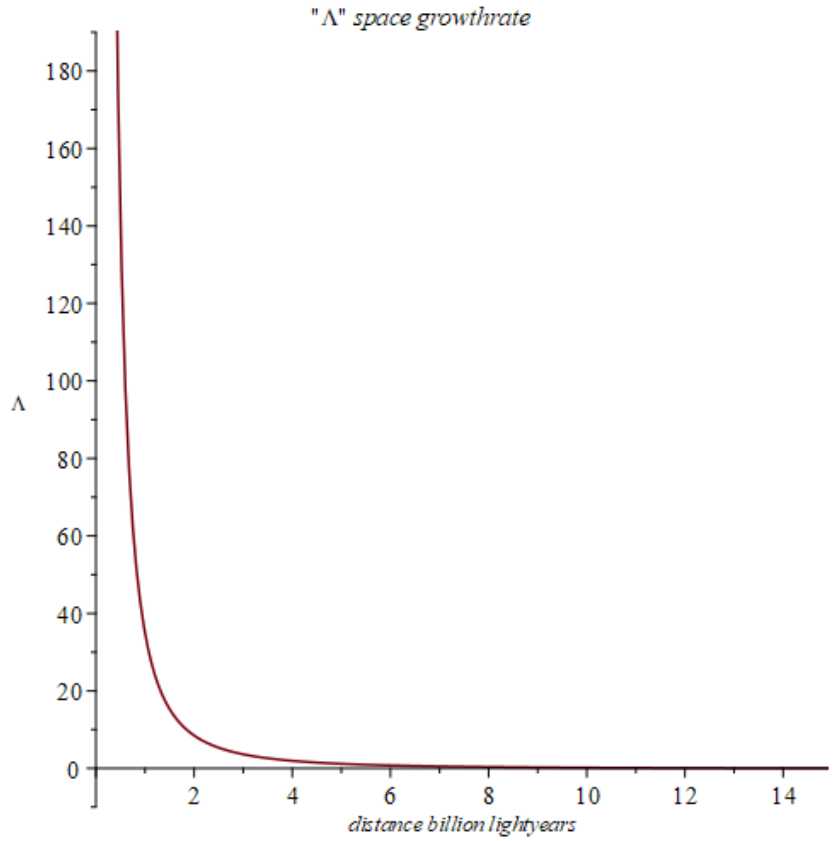
$$\frac{P(r,t)}{4\cdot\pi\cdot r^2} = \frac{(4\cdot R_s^2 - 2\cdot R_s\cdot r)\cdot c^2 + (-4\cdot R_s^2 + r^2)_c_2}{8\cdot\pi\cdot r^2\cdot c^2}$$

$\int_0^{2R_s}\frac{P(r,t)}{4\cdot\pi\cdot r^2}dr = \int_0^{2R_s}(\frac{(4\cdot R_s^2-2\cdot R_s\cdot r)\cdot c^2+(-4\cdot R_s^2+r^2)_c_2}{8\cdot\pi\cdot r^2\cdot c^2})dr = 2R_s$ the total space growth thus:

$_c_2 = -\frac{8\cdot c^2 R_s\cdot c^2}{7} + \frac{4}{7}c^2$ Thus, we have equation (7):

$$\Lambda flux = \frac{4\cdot R_s^2\cdot c^2 - 4(-\frac{8\cdot c^2 R_s\cdot c^2}{7} + \frac{4}{7}c^2)\cdot R_s^2}{8\cdot\pi\cdot c^2}\cdot\frac{1}{r^2} - \frac{R_s}{4\cdot\pi}\cdot\frac{1}{r} + \frac{-\frac{8\cdot c^2 R_s\cdot c^2}{7} + \frac{4}{7}c^2}{8\cdot\pi\cdot c^2}$$

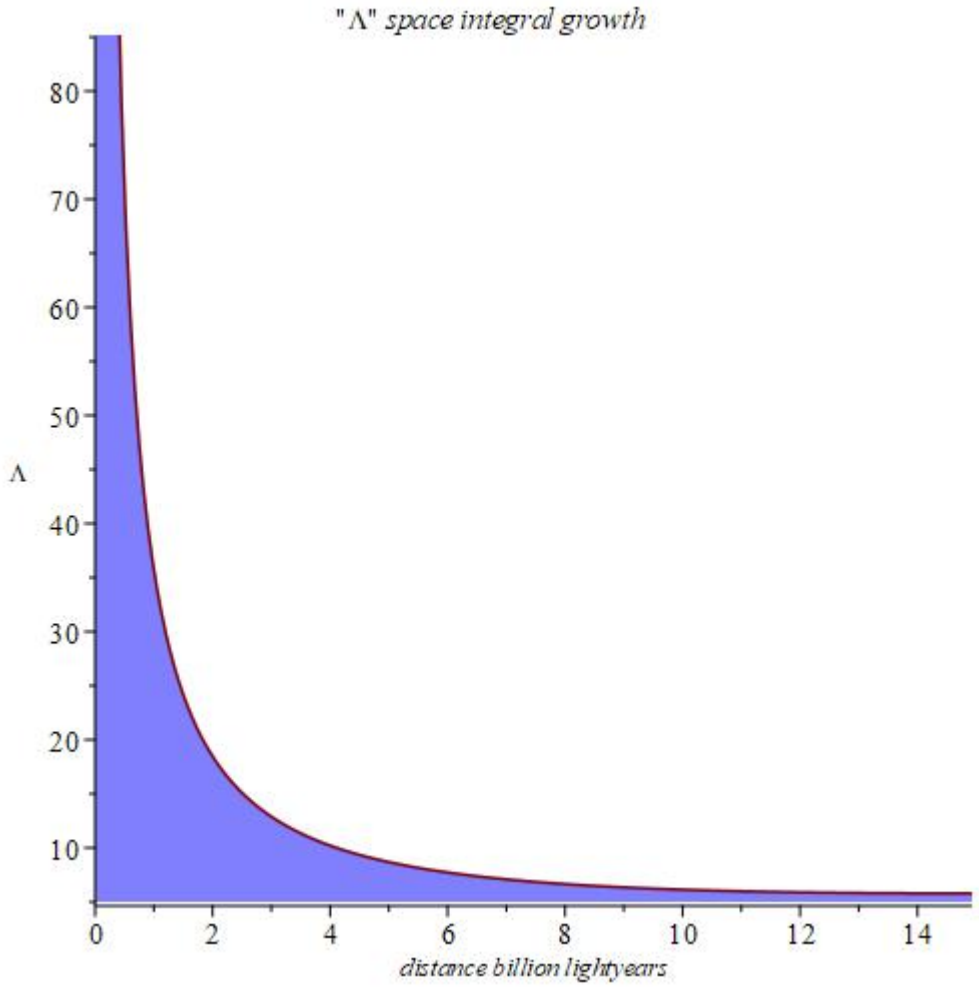
With the meaning of space flux growth rate for $R_s$ =7.46 and c = 3:



The redshift due to the space flux growth rate decreases parabolically as we can see from the Λ plot asymptotically to 0 at the radius of 7.464 Billion Light Years. After the first 4-5 billion light-years, the space flux rate is quite small and decreases to 0 for r = 2Rs.
Thus, the space flux growth decreases parabolically with radius and is equal to 0 at a diameter of 14.928 billion Light Years which is the border of our Local Universe. The space flux growth (8):

$$\Lambda = \frac{(2-4\pi)\cdot r^2 - 7\cdot R_s \cdot r - 16\cdot R_s^2 \cdot (\pi + \frac{3}{8})}{28\cdot \pi \cdot r^2}$$

Thus, for Rs=7.46, we have $\Lambda = \frac{35.6328}{r^2} - \frac{0.5939}{r} - 0.1201207$ and integral growth has the formula: -0.5939475112·ln(r)-0.1201207·r-35.6328/r where r in Billion light-years and the plot of positive growth:




After the Big Bang, there was inflation after that there was a quark epoch period of nucleosynthesis of hydrogen and helium-4 and probably Li when there was a very hot matter in expansion the universe cooled down. There were mostly hydrogen atoms and a sort of lower gravity through matter there was no time and no space: "A moment before the time at the place from out of space". The dark period ended in 362 million appear the First Star, thus the first day of the Local Universe took place after lower gravity produces the first huge star. And in the first nuclear reaction, there was a miracle that happened it generated light (photons) and Tau neutrino thus Time. We start from the moment that space thus μ neutrino was created 362 million years after the big bang which is when the first stars were born and the present-day thus the distance of comoving distance ~92 billion light-years.

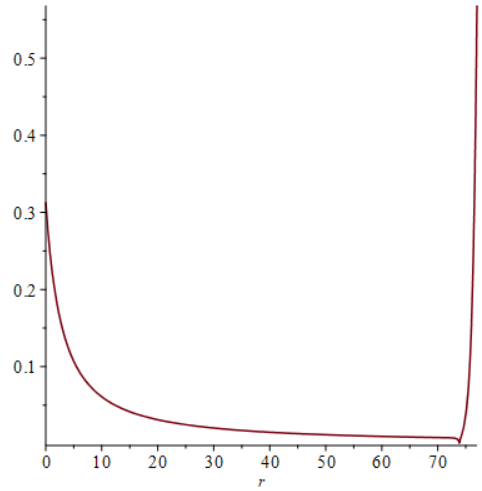
$$\int_{0.362}^{2\cdot R_S} \left(\frac{(2-4\pi)\cdot r^2 - 7\cdot R_s \cdot r - 16\cdot R_s^2 \cdot (\pi + \frac{3}{8})}{28\cdot \pi \cdot r^2}\right) dr = 92$$

After the distance of 7.15 million light-years between mass appears the rejection of gravitational forces that push galaxies further away then increase the void space between galaxies according to the onduscular equation that estimates the force. $\frac{R}{r}\Delta f(r,t) = \frac{1}{c^2}\frac{\partial^2 f(r,t)}{\partial t^2}$ (9)

The factor of R/r in the equation (3) interpreted as a wave equation says that the variable G propagates with the variable speed of $c \cdot \sqrt{\frac{R}{r}}$, but the gravitational waves travel at the constant speed of c.

Thus the Newton force $F_N$ is proportional to this variation thus according to (10):

$$\frac{r}{R}k = \frac{\frac{d^2 g_{(r)}}{dr^2}}{g(r)} \quad (10)$$ And we may extend to: $$F_N \sim \sqrt{\frac{\frac{\partial^2 \psi(r)}{\partial r^2}}{\psi(r)}} \quad (11)$$




The plot shows that the gravity force of repulsion grows after the value of 7.15 million lightyears.

The formula for redshift is $1+z = \lambda_{observed}/\lambda_{rest}$ thus we approximate z ~ v/c for v << c and the Doppler effect of expansion velocity that is the space is expanding: 72 kilometers per second per megaparsec.

We start with the supposition that the Local Universe exist for 13.79 billion years and the expansion of it is made with the light speed by muon neutrino (μ) thus the event horizon of the observable universe is 14.98billion and due to the expansion of space 92 billion lightyears for light to travel. We presume that the first stars appeared 362 million years after the Big Bang, and also appeared τ and μ neutrino thus time and space as we know it.

https://www.astro.ucla.edu/~wright/CosmoCalc.html

Thus, at the time when Galaxy GN-z11 and our galaxy were made should be around ~1.187 billion light-years at the time when the light was emitted, and the redshift shows that the space increased by a factor of 12.1 to 14.36 billion lightyears for light to travel and commoving radial distance 84.8 billion light-years as in the redshift.mw; With the parameters: $Omega_M$ = 0.018, $Omega_{vac}$ = 1, and Hubble constant $H_o$ = 69.6 we have the age at redshift z were 1.39billion years, commoving radial distance 82 billion light-years and the light travel time 24 billion lightyears.

The redshift and the name of the Galaxy GN-z11 are public at: https://en.wikipedia.org/wiki/GN-z11

30. Nov. 2022